Potential Harms Caused by AI Systems

Privacy Invasion in Social Media and Surveillance

https://mediaethicsmorning.wordpress.com/2015/02/25/blog-1-invasion-of-privacy/

Privacy Perception Across Cultures

Reuters

World Business Markets Sustainability Legal Breakingviews Technology Investigat

Just smile: In KFC China store, diners have new way to pay

By Reuters

September 1, 2017 3:21 PM GMT+5:30 · Updated 8 years ago

Aa

A customer tries Alipay's facial recognition payment solution "Smile to Pay" at KFC's new KPRO restaurant in Hangzhou, Zhejiang province, China September 1, 2017. REUTERS/Stringer Purchase Licensing Rights

Bits

Bits

Business, Innovation, Technology, Society

Germany's Complicated Relationship With Google Street View

BY CLAIRE CAIN MILLER AND KEVIN J. O'BRIEN APRIL 23, 2013 4:39 PM 3

Street View, which Google started in 2007, is in 50 countries, including Germany. Johannes Eisele/Agence France-Presse — Getty Images

How much people around the world think their data is worth?

In 2014, surveyed 900 people in five countries—the United States, the United Kingdom, Germany, China, and India. (Analytics and data science, Harvard Business Review.)

Do They Trust You with Their Data?

Percentages of consumers who said that each category of organization was "trustworthy" or "completely trustworthy" when it came to making sure that personal data was never misused.

87% Primary care doctors	**85%** Payment or credit card companies	**80%** E-commerce firms	**77%** Consumer electronics firms	**76%** Insurance companies	**76%** Banks
73% Telecom carriers	**70%** Technology firms	**68%** Internet giants (such as Google and Yahoo)	**66%** Governments	**61%** Media and entertainment companies	**56%** Social media firms

SOURCE TIMOTHY MOREY, THEODORE "THEO" FORBATH, AND ALLISON SCHOOP
FROM "CUSTOMER DATA: DESIGNING FOR TRANSPARENCY AND TRUST," MAY 2015

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Putting a Price on Data

Surveys of consumers in the United States, China, India, Great Britain, and Germany reveal that they value some types of information much more highly than others.

The approximate amount people say they would pay to protect each data type (per person, US$, 2014):

SOURCE TIMOTHY MOREY, THEODORE "THEO" FORBATH, AND ALLISON SCHOOP
FROM "CUSTOMER DATA: DESIGNING FOR TRANSPARENCY AND TRUST," MAY 2015

© HBR.ORG

https://hbr.org/2015/05/customer-data-designing-for-transparency-and-trust

Personally identifiable information (PII) or sensitive personal information (SPI)

- Full name (if not common)
- Home address
- Email address
- National identification number
- Passport number
- IP address
- Vehicle registration plate number
- Driver's license number
- Face, fingerprints, or handwriting
- Credit card numbers
- Date of birth
- Birthplace
- Genetic information
- Telephone number
- Login name, screen name, nickname, or handle

More Protected Than PII

- Medical records
- Children

Do We Really Need PII To Identify A Person?

- The absence of PII data does not mean that the remaining data does not identify individuals
- How many bits are needed to identify a person?
  - 8 billion people
    - Approximate gender → 4 billion
    - Approximate age, 3 buckets → 1.5 billion
    - Do they speak English? → 800M
    - Country? Occupation? Do they like hockey? → 10M
    - Chrome or Safari? Mac or Windows? Internet provider? → 1M

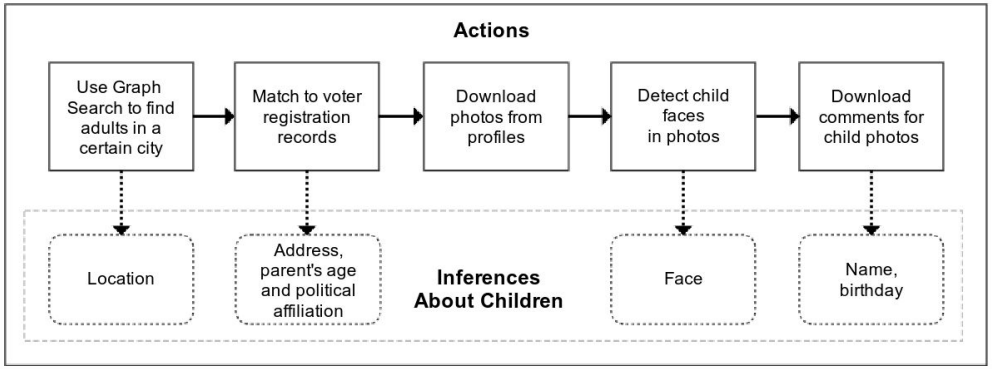
To encode 8 billion people we need around 35 yes/no questions.

For minorities you need much less, so these are more vulnerable!

Do People REALLY Care About Protecting Their Privacy?

- Facebook users in a suburban city on the East Coast in the USA.
- Each one of these users has been matched with high certainty to a particular registered voter in a voter registration list.

	Facebook	Instagram
Accounts		
Total collected	2,383	1,089
Sharing child photo	807	1,089
Sharing child name	365	689
Sharing child birthday	50	292
Photos		
Total collected	26,602	21,379
Child in photo	2,251	6,134
Name in comments	575	988
Birthday in comments	60	411



"Jerry", age 0: full name and birthdate known. His family consists of his father (name and birthdate known), mother (name known), and an older sister (name and birthdate known). The father's past and current occupations are known, and he is a football fan and political independent.

What about your parents'/siblings' profiles?

Minkus T, Liu K, Ross KW. Children seen but not heard: When parents compromise children's online privacy. InProceedings of the 24th international conference on World Wide Web 2015 May 18 (pp. 776-786).

What Can We Reveal Unknowingly?

From shopping habits at Target (in 2002), a mathematical models that could

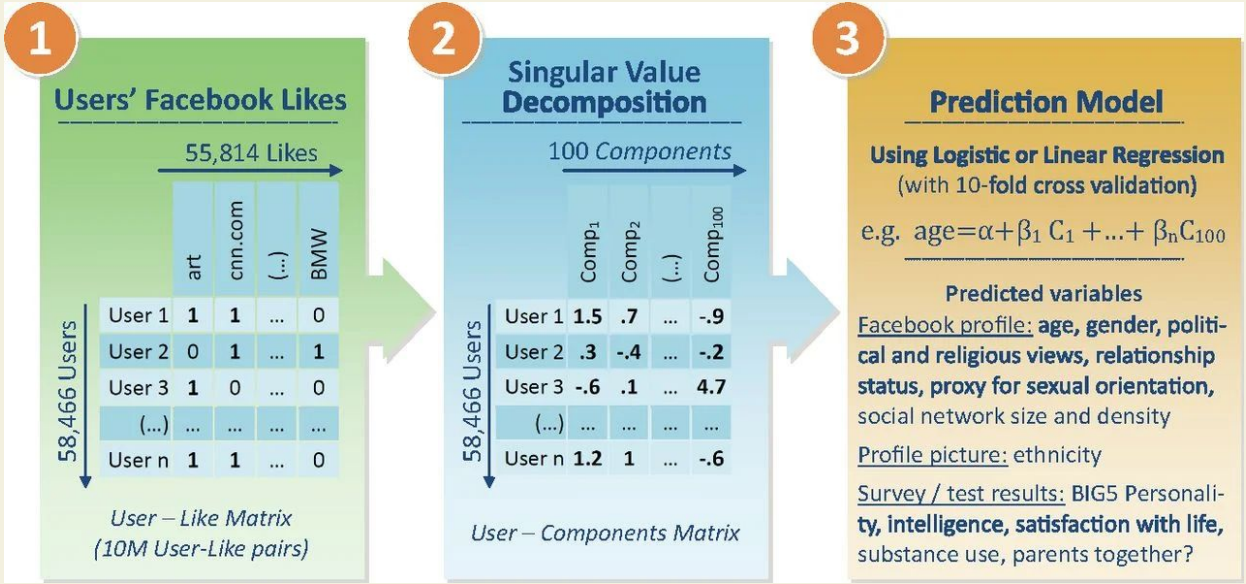
- determine which households contained kids, bachelors; which shoppers loved the outdoors and who was more interested in ice cream and which kind of novels.

“Facebook information (e.g., Likes), can be used to automatically and accurately predict a range of highly sensitive personal attributes including:

- sexual orientation, ethnicity, religious and political views, personality traits, intelligence, happiness, use of addictive substances, parental separation, age, and gender. ”

THE POWER OF HABIT,
https://ia803102.us.archive.org/35/items/CharlesDuhigg.ThePowerOfHabit_201808/Charles-Duhigg.The-Power-of-Habit.pdf

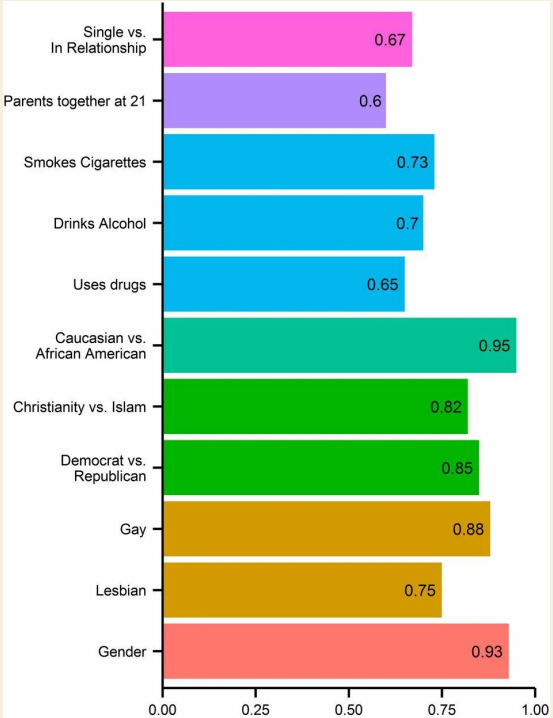
Kosinski M., Stillwell D., and Graepel T. (2013) Private traits and attributes are predictable from digital records of human behavior. *PNAS*



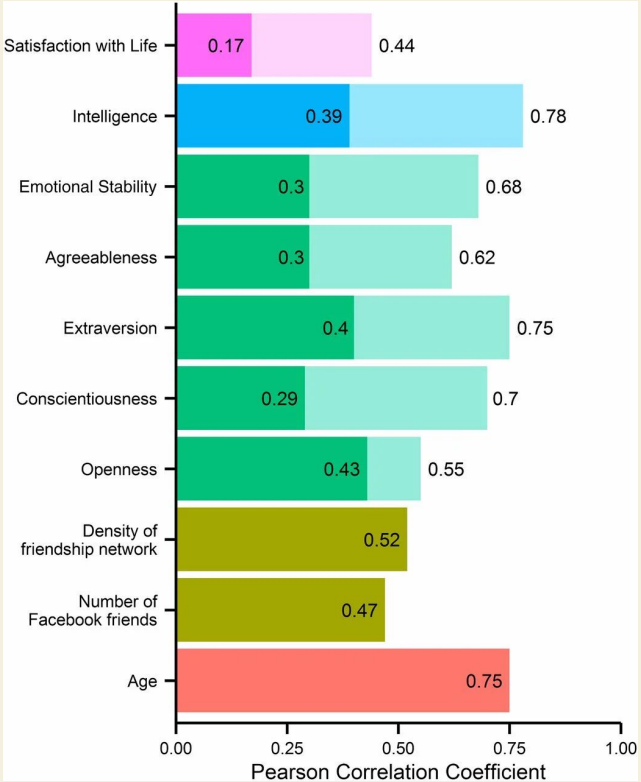
170 Likes per person on average Dimensionality Reduction

Facebook profile information obtained through the myPersonality Facebook application (www.mypersonality.org/wiki), a list of their Likes (n = 170 Likes per person on average), psychometric test scores, and survey information.

Kosinski M., Stillwell D., and Graepel T. (2013) Private traits and attributes are predictable from digital records of human behavior. *PNAS*



Prediction accuracy of classification for dichotomous/dichotomized attributes



Prediction accuracy of regression for numeric attributes and traits expressed

Kosinski M., Stillwell D., and Graepel T. (2013) Private traits and attributes are predictable from digital records of human behavior. *PNAS*

Cambridge Analytica

Political Exploitation (2016 US Presidential Election)

https://sites.google.com/michalkosinski.com/mypersonality/public

myPersonality.org

Home | Publications

PUBLICATIONS

SELECTED PUBLICATIONS BASED ON MYPERSONALITY DATA:

1. **Psychological targeting as an effective approach to digital mass persuasion** by S. C. Matz, M. Kosinski, G. Nave, and D. J. Stillwell, Proceedings of the National Academy of Sciences (PNAS), *2017.*
2. **Deep neural networks are more accurate than humans at detecting sexual orientation from facial images** by Y. Wang and M. Kosinski*, Journal of Personality and Social Psychology, 2018. (*Also see author notes; and preprint *Wang and Kosinski share the first authorship.*)
3. **Facial Width-to-Height Ratio Does Not Predict Self-Reported Behavioral Tendencies** by M. Kosinski, Psychological Science, 2017.
4. **Birds of a feather do flock together: behavior-based personality assessment method reveals personality similarity among couples and friends** by W. Youyou, D. Stillwell, A. Schwartz, & M. Kosinski*, Psychological Science, 2017. *(*Senior authorship.)*
5. **Mining Big Data to Extract Patterns and Predict Real-Life Outcomes** by M. Kosinski, Y. Wang, H. Lakkaraju, and J. Leskovec, Psychological Methods, 2016. (Sample dataset.)
6. **Computer-based personality judgments are more accurate than those made by humans**by W. Youyou, M. Kosinski*, D. Stillwell, Proceedings of the National Academy of Sciences (PNAS), 2015. *(*Youyou and Kosinski share the first authorship.)* ***The 19th most impactful paper of 2015.***
7. **Facebook as a Social Science Research Tool: Opportunities, Challenges, Ethical Considerations and Practical Guidelines** by M. Kosinski, S. Matz, S. Gosling, V. Popov, D. Stillwell, American Psychologist, 2015.
8. **Tracking the Digital Footprints of Personality** by R. Lambiotte and M. Kosinski, Proceedings of

Huge amounts of literature on predicting personal attributes of users.

Because these tasks have direct commercial use, to improve sales and recommender systems.

https://sites.google.com/michalkosinski.com/mypersonality/publications?authuser=0#h.p_dOt-Ut6bz1P4

Cambridge Analytica and Facebook Data Scandal

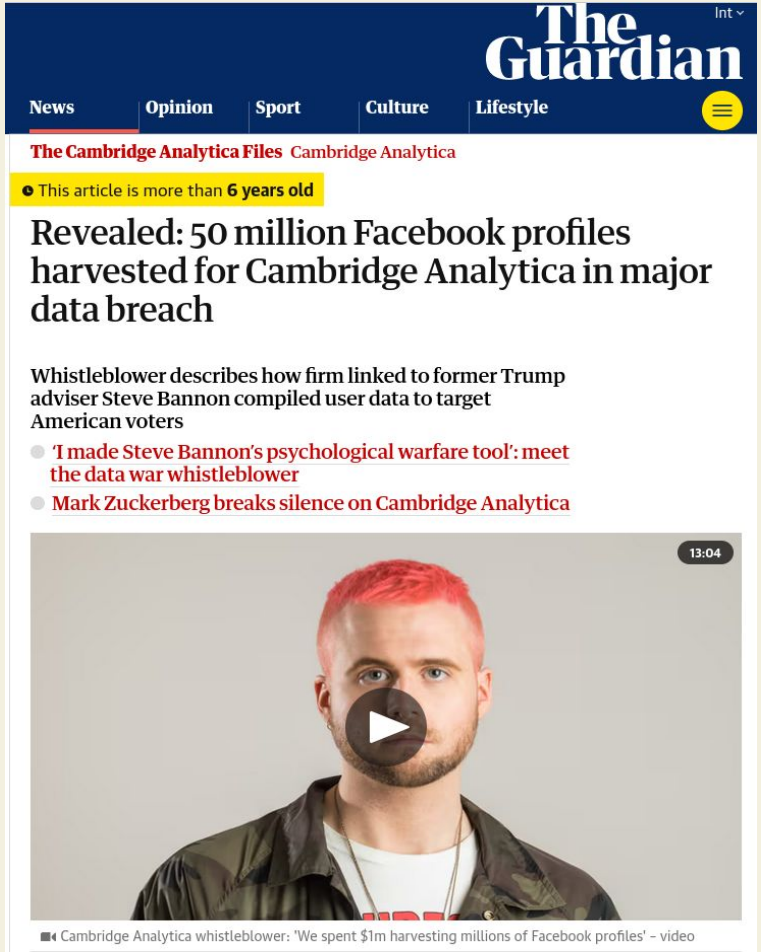
- In 2018, it became public knowledge that millions of Facebook users' data had been harvested without their consent.

- Key Players: Cambridge Analytica (CA) in partnership with Cambridge researcher, Aleksandr Kogan.

- Kogan created an app called "**this is your digital life**" featuring a personality quiz.

- CA paid for people to take it. agreed to have their data collected for academic purposes.

- The app collected:
  - Quiz results.
  - Personal data from users' Facebook accounts.
  - Data from users' Facebook **friends** (leading to **87 million profiles**).



The Guardian

Int

News | Opinion | Sport | Culture | Lifestyle

The Cambridge Analytica Files Cambridge Analytica

This article is more than **6 years old**

Revealed: 50 million Facebook profiles harvested for Cambridge Analytica in major data breach

Whistleblower describes how firm linked to former Trump adviser Steve Bannon compiled user data to target American voters

- 'I made Steve Bannon's psychological warfare tool': meet the data war whistleblower
- Mark Zuckerberg breaks silence on Cambridge Analytica

13:04

Cambridge Analytica whistleblower: 'We spent $1m harvesting millions of Facebook profiles' – video

Cambridge Analytica and Facebook Data Scandal

- The user had to have **a Facebook account and be a US voter** so that tens of millions of the profiles could be matched to electoral rolls.

- From an initial trial of 1000 test takers, the researchers collected **160,000 profiles or about 160 per person.**

- In a few months Kogan and CA had **a database of millions of US voters** which had its own algorithm to scan them, identifying possible political beliefs and personality characteristics.

- The paid test- takers had **agreed to share their data,** and Facebook's default terms <u>allowed their friends' data to be collected by an app</u>, unless they had changed their privacy settings.

- But **none of them had agreed to sell their data to firms such as Cambridge Analytica.**

Cambridge Analytica and Facebook Data Scandal

- Kogan passed the data to CA. CA used algorithms to psychologically profile people based on their Facebook activity.
- **Political Exploitation (2016 US Election)**: Donald Trump's campaign hired CA to target voters, where to make campaign stops and help with strategic communication, like what to say in speeches.
- CA used **"dark posts"** (personalized ads visible only to specific users) based on psychological profiles. Vulnerable users were influenced by pro-Trump messaging.
- In this case if the firm knew an individual was in favor of carrying firearms then he would be targeted with the following message

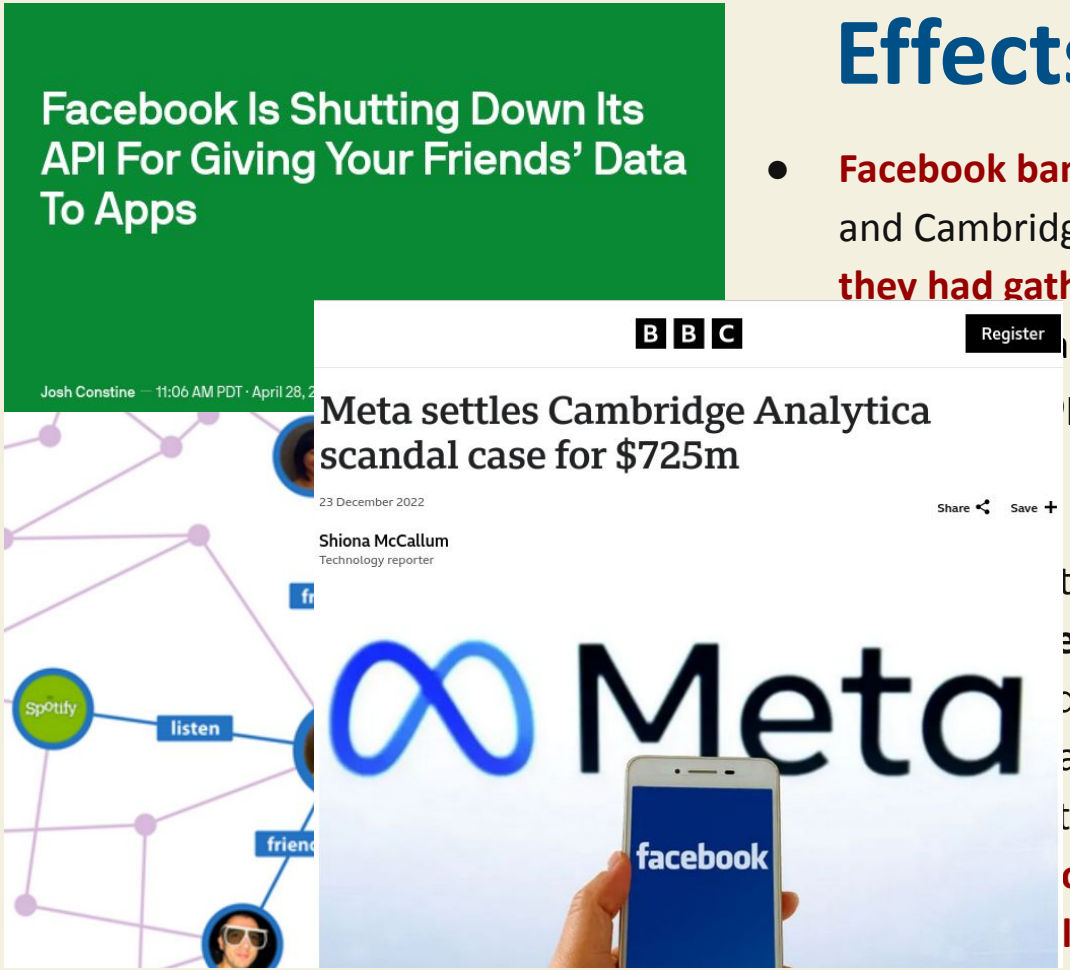
"Did you know Hillary Clinton wants to take your gun away?".

This **strategy contributed significantly to Trump's victory over Hillary Clinton.**

Who is responsible in the case of "Cambridge Analytica and Facebook Data Scandal?"

What measures could have been taken to avoid such an incident?

Effects of the Scandal



- **Facebook banned Kogan's app** and demanded Kogan and Cambridge Analytica **to destroy all the data that they had gathered** improperly.
- **ccess to data be removed if a user hasn't p in three months.**
- **in an app's access to data be limited** to profile photo and email address.
- **tion worldwide: Digital Services Act** and **ets Act** in the **European Union.**
- **ded to ban political ads all together; Google amount of targeting options; and Facebook the ability to opt out of political ads.**
- **o longer access your friend's data on less authorized by them.**

https://www.bbc.com/news/technology-64075067